ALLOMETRY ON ONE PAGE

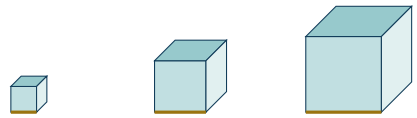
How big can a human be?

Stress, strain and a falling body: from the square–cube law to the quarter power of the mass–radius map.

1 / THE RULE AND THE PICTURE

Scale a body by λ and its forces outgrow its supports

Scaling a body by λ in every direction multiplies its weight by λ^3 and each supporting cross-section by λ^2, so the stress $\sigma = F/A$ (a pressure, in pascals) grows as λ, and with it the strain $\varepsilon = \Delta L/L = \sigma/E$ of a material of Young's modulus E: $\boxed{W \propto L^3,\ A \propto L^2 \ \Rightarrow\ \sigma = W/A \propto L}$



L $(1,1,1)$ $\quad$ $2L$ $(8,4,2)$ $\quad$ $3L$ $(27,9,3)$

quantity	expression	scales as
weight W	$\rho g L^3$	L^3
cross-section A	shape $\times L^2$	L^2
stress $\sigma = W/A$	$\rho g L$	L
hydrostatic pressure P	$\rho g h$	L
fall energy E_{fall}	$WL = \rho g L^4$	L^4
fracture energy E_{break}	ΓL^2	L^2

Γ is the fracture energy per area ($\mathrm{J\,m^{-2}}$); cubes show (W, A, σ).

2 / THE STATIC LIMIT IS GENEROUS

Weight alone would allow a column of kilometres

The stress at the foot of a column is $\rho g L$, which reaches the crushing stress σ_c at $L_s = \sigma_c/(\rho g)$. For bone, with $\sigma_c \approx 1.7 \times 10^8$ Pa and $\rho \approx 1900\ \mathrm{kg\,m^{-3}}$, this is about 9 km. A 70 kg person standing on two femora of about 3 $\mathrm{cm^2}$ each carries roughly 1 MPa, under one per cent of σ_c, so standing is not what limits a human, and the binding constraint must be dynamic.

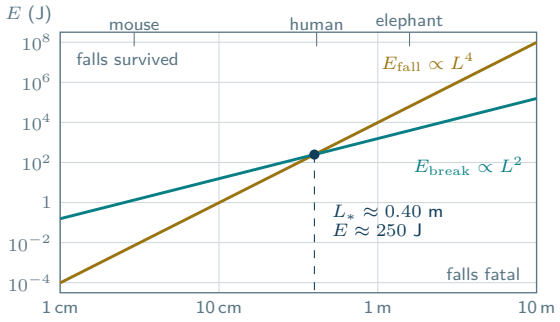
3 / A FALL AGAINST A FRACTURE

Energy in grows as L^4, energy absorbed as L^2

A fall through a height $\sim L$ releases $E_{\text{fall}} \sim WL = \rho g L^4$. A fracture crosses a plane of $(L/a_0)^2$ bonds, each worth $E_{\text{em}} = e^2/(4\pi\varepsilon_0 a_0) \approx 27$ eV, so $E_{\text{break}} = E_{\text{em}}(L/a_0)^2 = \Gamma L^2$ with $\Gamma = E_{\text{em}}/a_0^2 \approx 1.6\ \mathrm{kJ\,m^{-2}}$. In stress and strain terms the plane carries a force $\sigma_c L^2$, with $\sigma_c = E_{\text{em}}/a_0^3 \approx 3 \times 10^{13}$ Pa, through a rupture distance a_0, and work is force times distance.

4 / THE CROSSING AND THE QUARTER POWER

Equate the two energies and a limiting size appears



Setting $\rho g L^4 = \Gamma L^2$ gives $L_* = \sqrt{\Gamma/(\rho g)}$, which for water density and Earth gravity is 0.40 m and $M = \rho L_*^3 \approx 64$ kg; a mouse, human and elephant have mass-equivalent sides of 3 cm, 41 cm and 1.6 m.

In the units of the power graph, $\rho = m_p/a_0^3$, $\Gamma = E_{\text{em}}/a_0^2$ and the planetary ceiling supplies $g_P = GM_P/R_P^2 = Gm_pQ^{1/2}/a_0^2$ from $R_P = a_0 Q^{1/2}$ and $M_P = m_p Q^{3/2}$, so

$$\left(\frac{L}{a_0}\right)^2 = \frac{E_{\text{em}}}{m_p g_P a_0} = Q^{1/2}.$$

$$\frac{L}{a_0} \sim Q^{1/4}, \qquad \frac{M}{m_p} \sim Q^{3/4}, \qquad Q = \frac{e^2/4\pi\varepsilon_0}{Gm_p^2} \approx 1.2 \times 10^{36}$$

5 / WHY THE EXPONENT IS A QUARTER

The organism is the geometric mean of atom and planet

atom		organism		planet
a_0	$\xrightarrow{\times Q^{1/4}}$	$a_0 Q^{1/4}$	$\xrightarrow{\times Q^{1/4}}$	$a_0 Q^{1/2}$
0.53 Å	$\times 10^9$	5.6 cm	$\times 10^9$	6×10^7 m
m_p	$\xrightarrow{\times Q^{3/4}}$	$m_p Q^{3/4}$	$\xrightarrow{\times Q^{3/4}}$	$m_p Q^{3/2}$
1.7×10^{-27} kg	$\times 10^{27}$	2 kg	$\times 10^{27}$	2×10^{27} kg

Each step multiplies the length by $Q^{1/4}$, so $L^2 = a_0 R_P$. The static crushing length $\sigma_c/(\rho g_P) = a_0 Q^{1/2}$ is the planetary radius itself, since $\rho g_P R_P = e^2/(4\pi\varepsilon_0 a_0^4)$ is the atomic pressure, and the dynamic limit is the square root of atom times static length, $L = \sqrt{a_0 L_s}$.

6 / GRAVITY, MATERIAL AND SHAPE

What moves the limit, and what it leaves alone

Since $L_* = \sqrt{\Gamma/(\rho g)}$, the limiting size goes as $g^{-1/2}$ and the mass as $g^{-3/2}$, while the exponents stay put:

	g ($\mathrm{m\,s^{-2}}$)	L_*	M_*
Moon	1.6	$\times 2.5$	$\times 15$
Earth	9.8	$\times 1$	$\times 1$
Jupiter cloud tops	24.8	$\times 0.63$	$\times 0.25$

The bare powers give 5.6 cm and 2 kg because they use atomic density and $g_P \approx 44\ \mathrm{m\,s^{-2}}$; real chemistry and biomechanics, which Press's fuller estimate captures with a factor $\epsilon^{1/4}$, shift the prefactor and leave the quarter power.

ASIDE / PRESSURE IN A TALL BODY

Blood obeys $P = \rho g h \propto L$: head to foot of a 1.7 m human it is 17.7 kPa (133 mmHg), and a giraffe's brain, about 2 m above its heart, costs 156 mmHg, roughly why its arterial pressure is double ours. Long-bone diameters rise nearer $M^{3/8}$ than $M^{1/3}$, so large animals have stouter legs.

ASIDE / CHECK THE NUMBERS IN PYTHON

With `scipy.constants`, `Q = e**2/(4*pi*epsilon_0*G*m_p**2)` gives 1.24×10^{36}, and `a0*Q**0.25` and `m_p*Q**0.75` give 0.056 m and 1.96 kg. Replacing g_P by `9.80665` and ρ by 1000 in $\sqrt{\Gamma/(\rho g)}$ gives the 0.40 m above.

Convention. Section 5 of *The Architecture of Scale*, after Press (1980); $O(1)$ factors are dropped, so the powers locate a scale only.

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